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M.Tech.DEGREE EXAMINATION, DECEMBER 2024

First Semester

.21CSE584T - TIME SERIES ANALYSIS AND FORECASTING
(For the candidates admitted from the academic year 2024 - 2025)

Time: 3 hours

Max. Marks: 100

(5 x 20 = 100 Marks)
Answer **ANY FIVE** Questions

Marks BL CO PO

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| 1. (i) Explain the key differences between Univariate and Multivariate Time Series
(10 Marks) | 20 | 2 | 1 | 2 |
| (ii) Discuss the steps involved in Exploratory Data Analysis (10 Marks) | | | | |
| 2. (i) Build a program to perform Time series data visualization in python (10 Marks) | 20 | 3 | 1 | 2 |
| (ii) Explain output prediction model for a non-linearly separable data (10 Marks) | | | | |
| 3. Discuss the concepts of Moving Average and Simple Forecasting in Time Series Analysis.
Give an example to illustrate it. | 20 | 3 | 2 | 2 |
| 4. Explain exponential smoothing and weighted moving average with appropriate case study. | 20 | 3 | 3 | 2 |
| 5. Explain how multivariate data analysis is used in classification tasks. | 20 | 3 | 3 | 2 |
| 6. Implement ARIMA model in python and explain. | 20 | 4 | 4 | 2 |
| 7. Explain any one deep learning model for forecasting with an example. | 20 | 4 | 5 | 2 |
